CHALLENGE 3 · AUTONOMOUS ENTERPRISE

Mizan | ميزان

PDPL Compliance Agent

Reviews contracts and policies against Saudi Arabia's Personal Data Protection Law, clause by clause, with the evidence and the article behind every verdict.



Mays Alsalum · HUMAIN × Misk Future AI Builders

THE PROBLEM

PDPL compliance lives in documents, and reviewing them is slow, costly and specialist-bound.

Misses soft wording

Real violations read like normal business language, not like a rule being broken.

Over-flags

Compliant clauses that only look suspicious get flagged, and specialists stop trusting the tool.

Never says “unsure”

No evidence, no article, no escalation. In compliance, a confident wrong answer is worse than none.

These are the three ways a general-purpose AI fails at this task.

The PDPL has applied to every organisation processing personal data in the Kingdom since its grace period ended in September 2024.

THE SOLUTION

Mizan owns the first-pass review, end to end.

- Flags violations with the article and an exact quote
- Drafts a compliant rewrite, then re-checks it
- Sends low-confidence verdicts to a specialist

R06 · Violation · PDPL Art. 18 · confidence 0.95

"The Company shall retain all processed customer data indefinitely… regardless of whether the original processing purpose has expired."

Suggested fix · verified compliant

"The Company shall retain customer data only for as long as the processing purpose requires, and destroy it without undue delay afterwards."

Example clause from the synthetic development set; the fix shown is illustrative.

HOW IT WORKS

Six steps, three of them need no model at all.

1 · Code

Split into clauses

Follows numbering in sequence, Arabic or English; ignores page marks.

2 · Code

Gate

Skips clauses that never touch personal data, so the model reads only what matters.

3 · Model

Judge with evidence

Each clause against the rule catalog, with look-alike compliant examples to avoid false alarms.

4 · Code

Ground the quote

The evidence must exist in the clause. Mizan cannot cite text that is not there.

5 · Model

Fix and re-verify

Rewrites the clause, then judges the rewrite. Two failed tries: escalate.

6 · Decision

Report or escalate

Low confidence or “unclear” goes to a specialist with the reason, never a guess.

Rule catalog: 11 PDPL obligations from the official text, each with its article, a violation example and a compliant look-alike.

HOW WE TESTED IT

The right answer is known before the model writes a word.

Python plans each document: its type, language and which rules to break. The model only writes the text. A second model call checks every planted clause; failures are regenerated or dropped and logged.

Violations, half of them subtle

Traps: compliant clauses that look suspicious

Ambiguous: the right move is to escalate

Split	Docs	Use
dev	30	Development, error analysis
test	20	Locked; run once per frozen version
new type	5	Mobile app terms, never seen
noisy	5	PDF copy-paste noise
new rule	5	Hidden rule R08

65 documents, 60% Arabic and 40% English. A finding counts only if the rule is right and the quoted evidence points at the planted clause.

RESULTS ON 20 UNSEEN DOCUMENTS

Mizan missed no violation and cut false alarms on traps by 60%.

F1

0.952

Baseline 0.864

Recall, subtle violations included

1.000

Baseline 0.950

Traps wrongly flagged

7.7%

Baseline 19.2%

The trade-off: about 10× the cost, still under 2 US cents per document ($0.016 vs $0.0016), against an hour of a specialist’s time.

Baseline: one prompt, whole document, same model (gemini-3.5-flash-lite). Version frozen before the test run.

ROBUSTNESS AND ADAPTATION

A new obligation is one line of JSON, not a code change.

Hidden rule R08: cross-border transfer

0.29 → 1.00

F1 before and after adding R08 to the catalog. It was hidden during all development.

Condition	Baseline	Mizan
Test	0.864	0.952
New doc type	0.800	1.000
Noisy text	1.000	1.000

On the new document type the baseline crashed on 2 of 5 documents. Noise did not separate the two systems. Each robustness set has 5 documents: indicative, not conclusive.

EVALUATION AND ITERATION

Every change came from a measured failure.

Ver.	Change	Dev F1	What the evidence showed
v0	Baseline: one prompt	0.943	Strong, but 2 of 30 documents crashed
v1	Top-3 rules per clause	0.727*	Subtle wording dodged the keywords
v2	Gate + rules by doc type	0.913	All 6 misses: rule selection, not the judge
v3	Field-aware gate, all rules, context	0.990	Frozen, then run once on test

Ablation that changed the design

A second review of borderline verdicts sounded safer. With the same model as the judge it added false alarms (F1 0.913 with it, 0.931 without), so it is now off.

* v1 measured on 3 documents. Error analysis: eval/analyze.py names the cause of every missed violation.

WHAT IT CANNOT DO YET

Honest limits

- Escalation fires on low confidence, but caught none of the planted ambiguous clauses
- One model family wrote, filtered and reviewed the synthetic data
- 20 test documents, one run: indicative, not conclusive
- 11 obligations, not the whole law
- About 10× the cost of a single prompt

WHAT COMES NEXT

Next steps

- Escalate borderline clauses, not only low-confidence ones
- An independent model family for data and checking
- Validate on real documents with compliance specialists
- Larger test set, several runs, confidence intervals
- Route clear clauses to the cheapest model to cut cost

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Every verdict with its evidence and its article. Low confidence goes to a person.



github.com/MaysAlsalum/mizan-pdpl-agent

Reproduce every number for free: model calls are cached in the repository.

Mays Alsalum · Mizan assists a compliance specialist; it is not legal advice.